LITERATURE MAP · PREVIEW BUILD 21 SEPTEMBER 2026

# Condensates in twisted-bilayer optical lattices

Superfluidity across the moiré pocket crossover: how a weakly interacting condensate responds as the lowest band flattens

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**Abstract.** Twisted-bilayer optical lattices put two spin states of a Bose gas into square lattices rotated against each other and couple them with a microwave field, so that the Rabi coupling plays the part of interlayer tunnelling in twisted bilayer graphene. One experiment has been published, on $^{87}$Rb at a twist of 5.21°; it reports two superfluid regimes and a superfluid–insulator transition that moves with the coupling. This literature map collects what is known about how such a condensate stays superfluid as its lowest band flattens. It brings together the experiment and the lattice and continuum theory written about it, the theory of superfluid weight and quantum geometry developed for magic-angle graphene, recent work on condensates in flat bands, and the measurements of the superfluid fraction in modulated Bose gases. For a square bilayer with uniform coupling there is no magic angle: the band flattens because the atoms localise in the AA registry pockets, and the bosonic flat-band theory then rules out a geometric superfluid weight. The map argues that the open physics lies in the crossover, where band mass, density inhomogeneity and Josephson tunnelling between pockets compete, and it lists the calculations that would settle it. This preview contains the front pages and the context chapter.

**Keywords:** twisted-bilayer optical lattice; moiré superlattice; Bose–Einstein condensate; superfluid fraction; quantum metric; flat band; Rabi coupling; Josephson array

**At a glance: this preview build**

| 1 | 3 | 1 | 30 | 30 | 0 |
|---|---|---|---|---|---|
| chapters | pages of text | figures | sources | sources checked | open questions |

Planned for the full map: 10 chapters · 25 pages · 8 figures · 157 sources · 22 open questions

**Status:** preview build (front pages and chapter 3). Literature status 21 September 2026; checks of sources, numbers and quotations carried out on 21 September 2026 with AI assistance; see “How this map was made” on page i.

# The problem

Twisted-bilayer optical lattices bring moiré physics into a Bose gas. In the one experiment realised so far [1], two spin states of $^{87}$Rb sit in square lattices rotated by 5.21° and a microwave field couples them, in the role that interlayer tunnelling plays in twisted bilayer graphene. The experiment reports a moiré density pattern, two distinct superfluid regimes and a superfluid–insulator transition that shifts with the coupling. The question is how the superfluidity of a weakly interacting condensate evolves as the lowest band of such a lattice flattens.

It matters because it tests, for bosons and without disorder, strain or Coulomb interaction, how far the quantum-geometric picture of magic-angle graphene carries over, on the only moiré quantum gas with published data. It is hard because in a square bilayer with uniform coupling the band flattens by localisation into AA registry pockets rather than by interference at a magic angle. In the crossover window $\Omega/J \approx 5$–$25$, where the experiment operates, conventional ($m/m^*$), geometric (condensate metric), density-inhomogeneity (Leggett) and inter-pocket Josephson contributions to the superfluid weight compete, and inter-pocket tunnelling can fall below accessible temperatures.

Established: uniform coupling gives no magic angle and acts as a scalar moiré potential proportional to $\Omega$; a single flat band at a time-reversal-invariant momentum with uniform Bloch-state density cannot be superfluid for bosons within Bogoliubov theory; lattice models and Monte Carlo find clustered, glassy phases and thin superfluid domains destroyed at nK temperatures. Not yet found: Bogoliubov spectra, sound and the superfluid-fraction tensor of the Rabi-coupled condensate, its bosonic quantum geometry, finite temperature, and a trapped simulation compared with the data. Progress would be a quantitative account of superfluid fraction, sound and depletion across the crossover that reproduces the two superfluid regimes and the shifted transition.

# How this map was made

**Scope, method and conventions**

**Scope.** Bose gases in twisted-bilayer optical lattices (experiment, lattice and continuum theory), flat-band bosons and quantum geometry, measurements and definitions of the superfluid fraction, the electronic moiré context, numerical methods and the research landscape; fermionic and photonic analogues only where they bear on the bosonic question.

**Literature cutoff.** 21 September 2026. Preprints are included and marked as such; published versions are cited where they were found.

**How it was compiled.** The starting documents were five project documents and a two-page proposal naming about 100 sources. Web searches extended them on twisted-bilayer optical-lattice theory, flat-band bosons, superfluid-fraction measurements, electronic moiré, numerical methods and the research landscape, 2011–2026. Every source was then looked up in a primary record (arXiv or ar5iv, publisher page, code repository) in ten checking rounds with AI assistance: of the 30 sources cited in this build, 6 were confirmed or corrected, 24 partly verified and 0 not verifiable.

**Numbers.** Each number and quotation traces to a line of the project's facts ledger. Quotations are verbatim from the checked source; numbers confirmed only in part carry "as reported" or "as quoted".

**Conventions.** $E_R = h \times 3.67\,\mathrm{kHz}$ for $a = 395\,\mathrm{nm}$ [1]; $J$ single-layer nearest-neighbour hopping; $\Omega$ Rabi coupling ($\Omega_R$ in the experiment's notation); $L_M$ moiré period; $m$ commensurate index, $\tan(\theta/2) = 1/m$. The tag [computed for this map] marks numbers reproduced from the project's single-particle band-structure grid (no interactions, trap or temperature).

# In one page: the field and the argument

Moiré physics reached quantum gases in 2023, when a $^{87}$Rb condensate was loaded into two square lattices twisted by 5.21° and coupled by a microwave field [1]. The experiment found "two forms of superfluid and a modified superfluid to insulator transition". Magic-angle graphene suggests reading this through flat bands and quantum geometry [2–4]. We argue that for this lattice the reading has to change, in five steps.

1 **The lattice has no magic angle.** In twisted square bilayers flat bands need a spatially varying interlayer coupling with symmetry-enforced cancellation [5]; a uniform Rabi coupling instead acts on the lowest band as a scalar moiré potential proportional to $\Omega$ [6]. The band flattens because the atoms localise in the AA registry pockets: at $m = 8$, $V_0 = 8\,E_R$ the width falls to 0.25 of its uncoupled value at $\Omega/J \approx 16$ and to 0.004 at $\Omega/J \approx 32$, where the participation ratio is 0.008 and the Wannier spread 0.022 $L_M^2$ [computed for this map].

2 **Thesis 1: geometric superfluidity is excluded there.** For bosons, "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities.", and stable two-dimensional flat-band superfluidity "requires at least three bands within Bogoliubov theory" [7] (preprint). A band of isolated pockets is the atomic limit, where "no current is possible." [4]. The same authors conclude that "flat-band superfluidity is much more difficult to achieve than flat-band superconductivity." [7]

3 **The open physics is the crossover.** Between a dispersive band and isolated pockets, band mass, density inhomogeneity (the Leggett bounds [8, 9]) and tunnelling between pockets all carry superfluid weight. No calculation of this competition for the Rabi-coupled condensate was found in the literature checked for this map. The tools to decide it exist in simpler geometries: sound anisotropy and Leggett bounds measured in modulated two-dimensional gases [10–12], and Josephson oscillations between the cells of a supersolid [13].

4 **Thesis 2, working hypothesis: SF-II is a Josephson array of AA pockets.** The experiment's second regime is described as "superfluid domains embedded in a gapped insulator" [1]. At $V_0 = 8\,E_R$ its moiré-visibility threshold $\Omega_R > 0.5\,E_R$ sits at $\Omega/J \approx 16$, where the lowest band detaches and has lost 75 % of its width [computed for this map]. The hypothesis fails if the superfluid fraction exceeds the Leggett prediction for the pocket density, if sound scales as $U$ (the flat-band signature [14]), or if the threshold does not track $\Omega/J$ as $V_0$ changes.

5 **Progress** would be a two-component Gross–Pitaevskii calculation of the superfluid-fraction tensor, sound and depletion across the crossover, with Leggett brackets and a trap, that reproduces the two superfluid regimes and the shifted transition [1, 10, 12]. A finite-temperature treatment would show whether the inter-pocket tunnelling survives at the temperatures of the experiment, which the paper, as read for this map, does not state.

**Where quantum geometry still enters**

The geometric formulas do not disappear from the problem. The condensate metric at the condensation momentum sets sound and stability in the bosonic theory [7, 14], so it is a diagnostic of how far the crossover has progressed. Band touchings [15] and detuned couplings, which map onto Hofstadter-type spectra [16], are the places where a genuinely geometric window could still open. Chapter 3 explains these ideas for readers from the Bose-gas side.

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## Chapters planned for the full map

**1 The topic in brief** planned, 3 pp.
*The platform, the one experiment and its four regimes, and the variants.*

**2 Foundations** planned, 3 pp.
*Supercells, the Rabi-coupled GPE, pocket formation and superfluid fraction.*

**4 Data, benchmarks and verification** planned, 2 pp.
*How phases and superfluid fractions are measured, and cross-method checks.*

**5 State of the art** planned, 4 pp.
*The experiment and every theory study, with a coverage chart and timeline.*

**6 Position** planned, 3 pp.
*Geometric superfluidity is excluded; the pocket crossover is the open problem.*

**7 Around the topic** planned, 2 pp.
*Photonic moiré, quasicrystals, dipolar bilayers and coupled 1-D gases.*

**8 Research and industry perspective** planned, 3 pp.
*Groups, open codes, compute, funding and flat-band superconductivity.*

**9 Open questions** planned, 3 pp.
*Twenty-two questions in five themes, and directions not yet addressed.*

**10 Reading paths and outlook** planned, 1 p.
*Where to start, by background, and what comes next.*

### How to read this map

**Twenty minutes.** The problem, the one-page summary, then the key-points boxes at the end of each chapter.

**New to electronic moiré and quantum geometry.** Chapter 3 (in this preview), then chapter 6 (planned).

**Looking for a research question.** Chapters 6 and 9 (planned).

Boxes: key points, what a result means for the reader, open questions. Numbers tagged [computed for this map] come from the project's own band-structure grid.

# 3 Context: electronic moiré and quantum geometry for Bose-gas physicists

*How flat moiré bands in graphene become superconducting through quantum geometry, and which parts of that account survive for a condensate in a twisted optical lattice.*

We borrow most of the vocabulary for twisted-bilayer optical lattices (TBOL) from twisted bilayer graphene (TBG). This chapter covers how a moiré band becomes flat, why a flat band can still carry superfluid weight, and what changes for a condensate. Figure 3.1 traces the argument; Table 3.1 translates the terms.

## 3.1 Flat bands by interference: the magic angle

In the Bistritzer–MacDonald (BM) continuum model the Dirac cones of the two graphene layers are coupled by an interlayer tunnelling $w$ that varies with the moiré pattern [2]. One dimensionless number controls the low-energy bands, $\alpha = w/(vk_\theta)$, where $v$ is the Dirac velocity and $k_\theta$ the moiré wavevector. The renormalised velocity is $v^*/v = (1-3\alpha^2)/(1+6\alpha^2)$, and in the authors' words, "For a discrete set of magic angles the velocity vanishes, the lowest moire band flattens". The first magic angle is $\theta \approx 1.05°$. Nothing localises here. Tunnelling paths interfere destructively while the states stay extended over the moiré cell, and this is exactly where the optical lattice will differ.

Experiments then found correlated insulators at half filling of the flat bands [17] and superconductivity at "twist angles of about 1.1°—the first 'magic' angle", with "a critical temperature of up to 1.7 kelvin" at "a carrier density of about $10^{11}$ per square centimetre" [3, 18].

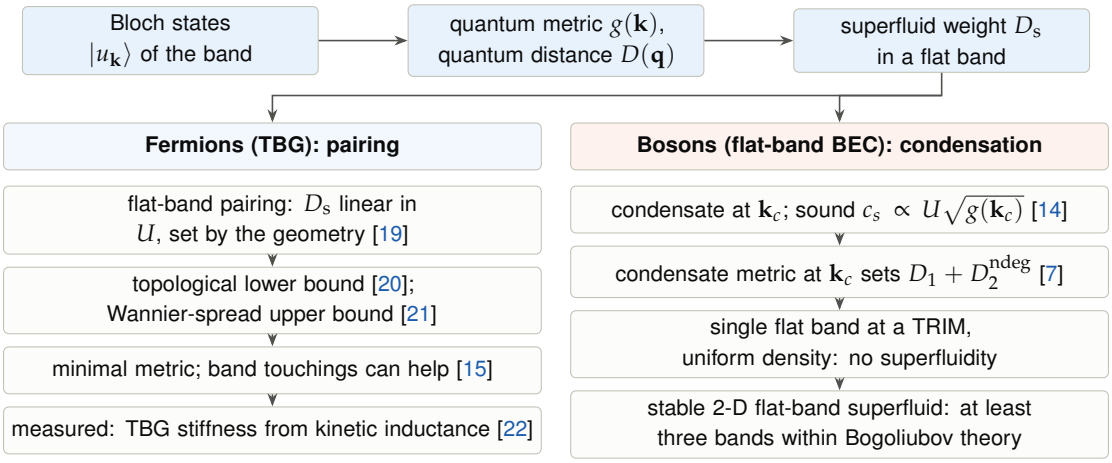



**Figure 3.1:** From Bloch states to superfluid weight: the shared trunk (top), then the fermion branch (pairing) and the boson branch (condensation at $\mathbf{k}_c$; [14] and the preprint [7]).

## 3.2 Superfluid weight from quantum geometry

In a dispersive band $D_s \propto n/m^*$, so a flat band with $m^* \to \infty$ looks like a dead end. It is not. The quantum metric $g(\mathbf{k})$ measures how fast the Bloch state $|u_\mathbf{k}\rangle$ changes with $\mathbf{k}$, and the quantum distance $1 - |\langle u_\mathbf{k}|u_{\mathbf{k}'}\rangle|^2$ is its finite-step version. Peotta and Törmä showed that pairing in a flat band still produces a superfluid weight, linear in the attraction ("$D_s \propto U$ in a flat band", as the abstract puts it) and controlled by this geometry [19]. For TBG the superfluid weight has "a topological lower bound proportional to the pairing gap" [20]. From the other side, Verma, Hazra and Randeria bound the low-energy optical weight, and so the stiffness, of a trivial flat

band by the Wannier spread [21]: well-localised orbitals mean little stiffness. Huhtinen *et al.* then showed that the physically relevant quantity is the minimal quantum metric, which "is related to the superfluid weight in isolated flat bands", and that "a non-isolated flat band can actually be beneficial" [15]. The review [4] states the limiting case plainly: "a trivial flat band is equivalent to an insulator in the atomic limit of large atom-atom separation", and there "no current is possible."

The stiffness of magic-angle TBG has now been measured. Tanaka *et al.* combined "d.c. transport and microwave circuit quantum electrodynamics to directly measure the superfluid stiffness of superconducting MATBG through its kinetic inductance" and found it "much larger than expected from conventional Fermi liquid theory" [22]; a companion measurement did the same in twisted trilayer graphene [23, 24]. A BM-model decomposition attributes 22–26 % of $D_s$ at charge neutrality to quantum geometry [25] (preprint).

### 3.3 Condensates in flat bands

A condensate has no pairing gap. It occupies one Bloch state at momentum $\mathbf{k}_c$, and in a flat band the interaction energy $Un_0$ is the only scale. Julku, Bruun and Törmä worked this out for the kagome lattice [14]. The projected Bogoliubov spectrum is

$$E_1(\mathbf{k}_c+\mathbf{q}) = \frac{Un_0}{3} D(\mathbf{q}), \qquad D(\mathbf{q}) = \sqrt{1-|\langle u_1(\mathbf{k}_c+\mathbf{q})|u_1(\mathbf{k}_c-\mathbf{q})\rangle|^2}, \tag{1}$$

so the excitations are set by the quantum distance, the sound velocity $c_s = (2Un_0/3)\sqrt{g(\mathbf{k}_c)}$ is linear in $U$ rather than in $\sqrt{U}$, and the depletion as $U \to 0$ "is determined by the quantum geometry only". The companion general theory [26] assumes, as the project documents summarise it (the paper was not read for this map), a uniform condensate density over the orbitals, an isolated band and projected interactions. Julku, Salerno and Törmä added "orbital-position-independent generalizations of the quantum metric" and showed that $D_s$ must be taken at fixed total particle number [27].

The 2026 preprint by Huhtinen *et al.* sharpens the picture [7]. For uniform-density condensates at weak coupling, $D_1 + D_2^{\mathrm{ndeg}} = (2Un_0^2/M)\,\mathcal{M}^D(\mathbf{k}_c)$, where $\mathcal{M}^D$ is the condensate quantum metric: a local quantity at $\mathbf{k}_c$, not an integral over the Brillouin zone (BZ). The global geometry enters the fluctuation term $D_3'$, which in their model "is positive only at very small interactions, and turns negative around $U \approx 0.13t$". At a time-reversal-invariant momentum (TRIM), "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities." And a large integrated metric does not help: on the checkerboard lattice, "even though the integrated quantum metric diverges due to the band touching, condensation and superfluidity remain impossible." The authors conclude that stable two-dimensional flat-band superfluidity "requires at least three bands within Bogoliubov theory" and that "flat-band superfluidity is much more difficult to achieve than flat-band superconductivity." A companion preprint phrases the criterion in real space: "frameworks consisting of triangles with nonzero area are promising for condensation, whereas … square frameworks indicate condensation in a single mode is impossible" [28].

Two further preprints add to this. For a two-band condensate, "in the case of U(2) invariant interactions, the superfluid density is zero" [29]; the TBOL experiment is modelled with SU(2)-symmetric interactions $g_{11} = g_{22} = g_{12}$ [1], but no source checked tests the result there. Burgher *et al.* relate the sound velocity in quasi-flat bands to a "quantum geometric kernel" [30].

**Table 3.1:** Dictionary between twisted bilayer graphene and twisted-bilayer optical lattices.

| TBG | TBOL | What it means |
|---|---|---|
| layer | spin state, each with its own lattice [1] | the "layers" are internal states, not planes in space |
| interlayer tunnelling $w$ | Rabi coupling $\Omega$ (microwave) | uniform in space for a microwave drive |
| twist angle, magic angle | twist $\theta$, index $m$; no magic angle for uniform $\Omega$ [5] | flattening is set by $\Omega/J$, not by $\theta$ |
| AA / AB stacking | AA / AB registry regions | where the two lattices' sites coincide or alternate |
| BM continuum model | scalar moiré potential $\propto \Omega$ [6] | one effective band in a moiré potential |
| superfluid weight $D_s$ | superfluid fraction $f_s$ (a tensor) | response to a phase twist |
| quantum metric | condensate quantum metric at $\mathbf{k}_c$ [7] | local geometry, not the BZ integral |
| Cooper pairs | condensate in one Bloch state | what the superfluid is made of |
| Coulomb interaction | contact interaction $g$ | short range, tunable, no screening |
| transport, kinetic inductance [22] | sound velocity, phase twist, Leggett bounds [10, 11] | how the stiffness is measured |

## 3.4 What carries over, and what does not

Both systems ask what carries superfluid weight once the kinetic energy of a moiré band is gone. TBG samples come with twist-angle disorder, strain and long-range Coulomb repulsion, and the superfluid is made of pairs. The optical lattice is clean, the interaction is a contact term, and the superfluid is one macroscopic state whose density can be imaged; its superfluid fraction follows from sound anisotropy, $f_{s,x} = c_x^2/c_y^2$, and from Leggett's bounds on that density [10–12].

The real difference lies in how the band flattens. Flat bands in twisted square bilayers need a spatially varying interlayer coupling whose symmetry forces exact cancellations [5]. A microwave Rabi coupling is uniform, and projected on the lowest band it acts as a scalar moiré potential of amplitude proportional to $\Omega$ (derived for 3-D quasi-moiré crystals and applied to 2-D by the project) [6]. The band flattens by localisation in the AA pockets, not by interference, and at $m = 8$, $V_0 = 8\,E_R$ the coarse-grained AA/AB density ratio is already 110 at $\Omega/J \approx 16$ [computed for this map]. The bosonic geometric results assume the opposite, a uniform density.

**What this means for Bose-gas physicists**

A flat-band condensate has a clear signature: sound linear in $U$ and a depletion that stays finite as $U \to 0$ [14]. The metric that counts is the local one at $\mathbf{k}_c$, not the BZ integral of the fermionic literature. In a square lattice with uniform coupling the flat limit is the atomic limit, so the geometric formulas become diagnostics of the crossover, not a mechanism.

**Open**

No calculation of the condensate metric of the Rabi-coupled bands was found for this map. Open: what the uniform-density results say for AA pockets, whether [29] transfers, and how the geometric term compares with the Leggett bound.

**Key points**

- TBG flattens by interference, a uniform-$\Omega$ TBOL by localisation [2, 5, 6].
- Bosons: one flat band at a TRIM with uniform density is not superfluid [7].
- AA pockets break uniform density: the open physics is in the crossover.

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# A Glossary

*Terms from cold-atom physics, electronic moiré materials and quantum geometry, as this map uses them.*

**Moiré pattern.** The long-wavelength beat between two periodic structures that are rotated or stretched against each other; its period $L_M$ is much larger than the lattice constant at small twist.

**Commensurate supercell.** The common superlattice of two lattices at a commensurate twist, for square lattices $\tan(\theta/2) = 1/m$ with integer $m$; it makes band calculations periodic.

**Registry (AA, AB).** How the sites of the two layers line up locally: on top of each other in AA regions, alternating in AB regions. AA regions become the pockets where atoms collect.

**Rabi coupling.** A coherent drive (microwave or two-photon Raman) that converts atoms between two internal states; in a twisted-bilayer optical lattice it plays the role of interlayer tunnelling. Written $\Omega$.

**Tune-out wavelength.** A laser wavelength at which the light shift of one atomic state vanishes, so that a lattice at that wavelength traps only the other state; used to give each spin state its own lattice.

**Recoil energy.** $E_R = \hbar^2 k^2/2m$ for the lattice wavevector $k$; the natural energy unit of optical lattices. Lattice depths $V_0$ are quoted in $E_R$.

**Bose–Hubbard model.** Lattice model of bosons with nearest-neighbour hopping $J$ and on-site repulsion $U$; valid when the lowest band is deep and well separated.

**Gutzwiller approximation.** Mean-field product of single-site states; captures the Mott transition but no inter-site correlations.

**Cluster Gutzwiller.** Gutzwiller theory with small clusters of sites treated exactly, which restores short-range correlations.

**Worm quantum Monte Carlo.** Path-integral Monte Carlo that samples open world-line ends; exact for bosons without a sign problem and gives the superfluid fraction from winding numbers.

**TWA and SPGPE.** Truncated Wigner approximation: classical fields with quantum noise in the initial state. Stochastic projected GPE: a classical-field theory coupled to a thermal reservoir.

**GPE.** Gross–Pitaevskii equation: the mean-field equation for the condensate wavefunction; here with two components and a Rabi term.

**BdG (Bogoliubov–de Gennes).** Linearisation around a mean-field state; for a condensate it gives the Bogoliubov excitation spectrum, sound velocity and depletion.

**Superfluid fraction.** $f_s$: the part of the fluid that does not follow a slowly moving boundary, or equivalently the response to a phase twist; a tensor in anisotropic potentials.

**Superfluid weight.** $D_s$: the coefficient of the energy cost of a phase twist (superfluid stiffness); $f_s$ is $D_s$ normalised by its free-particle value.

**Leggett bounds.** Upper and lower bounds on $f_s$ computed from the density profile alone; the upper bound is exact for a one-dimensional modulation of a weakly interacting condensate.

**Quantum metric.** $g_{\mu\nu}(\mathbf{k})$: the real part of the quantum geometric tensor; it measures how fast the Bloch state changes with $\mathbf{k}$.

**Condensate quantum metric.** The metric evaluated at the condensation momentum $\mathbf{k}_c$ in the basis of the condensate; it sets sound and stability of a flat-band condensate.

**Minimal quantum metric.** The integrated metric in the orbital basis that makes its diagonal components smallest; the quantity tied to the superfluid weight of paired fermions in an isolated flat band.

**Quantum distance.** $1 - |\langle u_{\mathbf{k}} | u_{\mathbf{k}'} \rangle|^2$ between two Bloch states; its small-step limit is the quantum metric.

**Wannier function and spread.** A localised orbital built from the Bloch states of a band; the spread is its mean-square size. A small spread means a band of nearly isolated orbitals.

**Participation ratio.** A measure of how many sites or cells a density spreads over, normalised to one for a uniform density; it falls towards zero as atoms collect in pockets.

**TRIM.** Time-reversal-invariant momentum: a point of the Brillouin zone equal to its negative up to a reciprocal-lattice vector ($\Gamma$, $X$, $M$ in a square lattice).

**BM model.** Bistritzer–MacDonald continuum model of twisted bilayer graphene: Dirac cones of the two layers coupled by moiré-periodic tunnelling.

**Magic angle.** A twist angle at which the Dirac velocity of the BM model vanishes and the lowest moiré band flattens; the first lies near $1.05^\circ$ [2].

**Flat band.** A Bloch band whose width is small compared with the interaction energy, so that interactions decide the ground state.

**FCI / FQAH.** Fractional Chern insulator, fractional quantum anomalous Hall state: fractional quantum Hall physics in a flat Chern band without an external magnetic field.

**Bose glass.** An insulating but compressible phase of bosons in a disordered or quasiperiodic potential, with local superfluid puddles but no global phase coherence.

**Josephson array.** A lattice of superfluid islands coupled by tunnelling; its superfluid weight is set by the inter-island Josephson coupling.